

Lend-Sight Mobile: Mobile Sales Data Analysis

An enterprise-grade interactive Business Intelligence Dashboard validating complete market dynamics, brand revenue, demographics, and transactional behaviors using Power BI & SQL.



PRESENTED BY
Business Intelligence Unit



CORE ARCHITECTURE
Full-Stack Analytics Pipeline



TARGET ENVIRONMENT
Power BI Desktop & Cloud

Project Scope & Objectives

INTRODUCTION



Brand & Model Performance

Evaluate unit volumes and revenue patterns of manufacturing giants (Apple, Samsung, etc.) against dynamic model releases.



Demographic Segmentation

Break down buying power by dividing users into specific age cohorts, detecting purchasing habits, preferences, and ratings.



Geographical Insights

Establish high-density revenue hubs utilizing map plotting, tracking city performance, pricing tolerance, and regional demand.



Transactional Behavior

Correlate user choices in payments (Credit Cards, UPI, Cash) with specific product pricing levels and rating responses.

Tools & Technological Stack

ARCHITECTURE



MySQL Engine

Powers foundational transactional data pipeline.
Facilitated initial data validation queries,
structural schema verification, and primary key
constraints mapping.



Power Query / Python

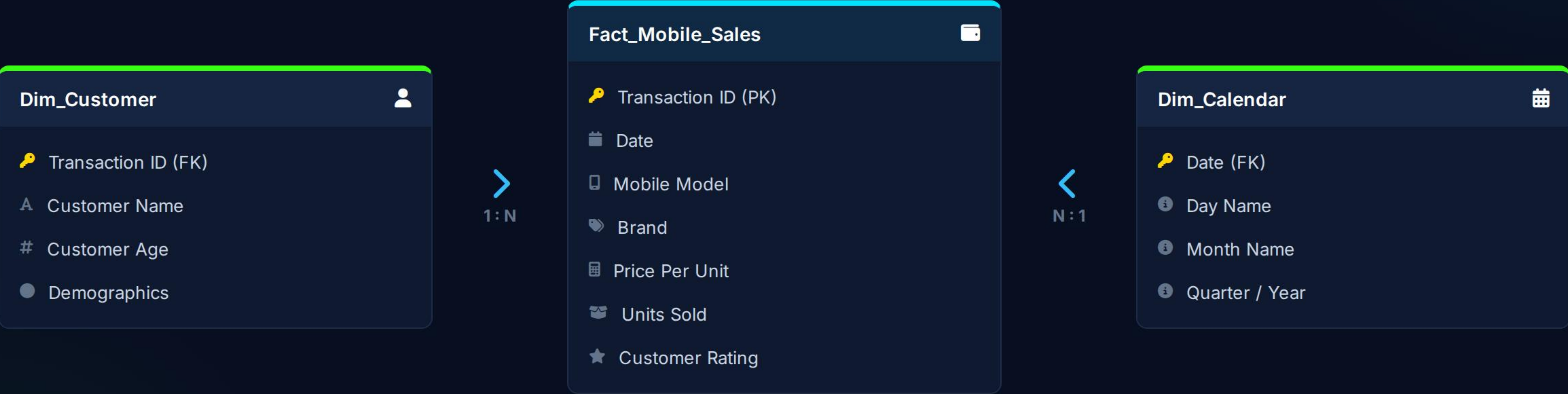
Designed ETL pipeline directly inside Power BI
and Jupyter Notebooks. Executed
STR_TO_DATE functions, standardized strings,
and calculated age buckets.



Power BI & DAX

Constructed highly optimized semantic model.
Implemented custom DAX variables, time-
intelligence calendar dimensions, and dynamic
real-time reporting cards.

Data Schema & Modeling



Preprocessing & Validation Flow

METHODOLOGY



1. De-duplication

Identified and removed redundant records utilizing the primary key constraint on Unique Transaction ID.



2. Format Alignment

Converted text-based date markers to standard ISO formats using SQL STR_TO_DATE for robust time intelligence analysis.



3. Outlier Filtration

Cleaned numeric values, filtering erroneous customer ratings, zero values in sales units, and empty customer names.



4. DAX Integration

Built DAX metrics to bridge tables and generate age groups, sales margins, and dynamic growth percentages.

Dashboard 1: Sales & Revenue Summary

POWER BI LIVE VIEW



REGIONAL SALES OVERVIEW

BY TOP CITIES

City Hub	Total Units	Total Revenue	Avg Rating
Hyderabad	12,455	\$152,430,000	4.35 ★
Mumbai	10,120	\$124,199,000	4.20 ★
Bangalore	8,145	\$99,158,000	4.15 ★
Chennai	5,120	\$62,300,000	3.98 ★

Dashboard 2: Trends & Market Penetration

VISUAL OVERVIEW

MONTHLY REVENUE TREND ANALYSIS

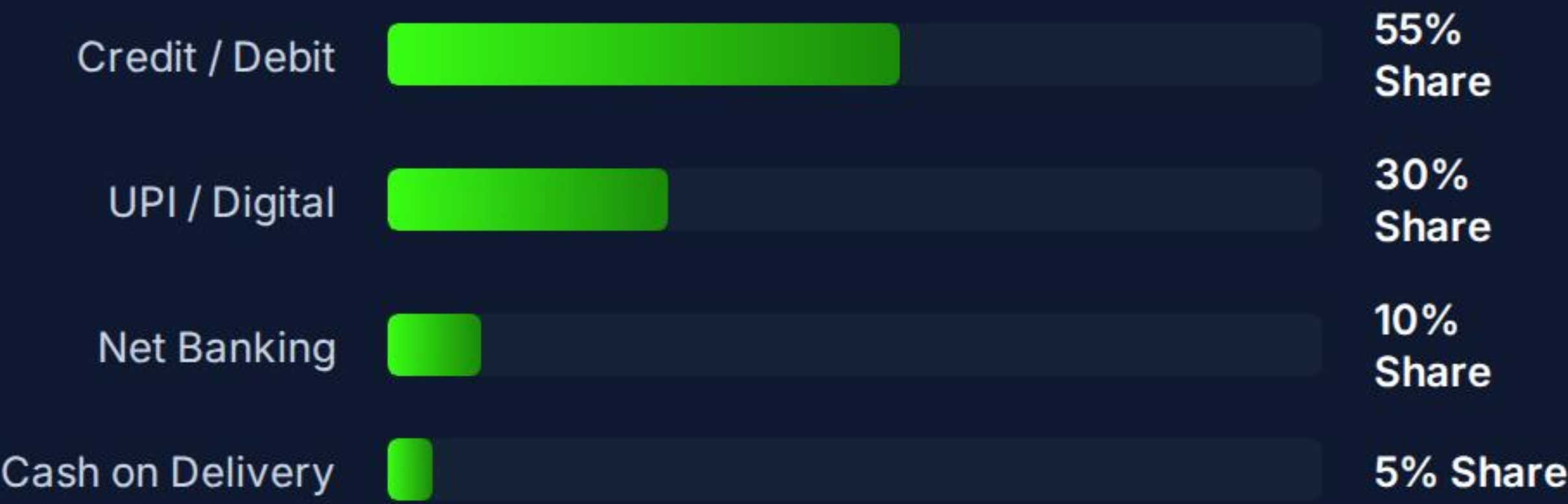
QUARTER-OVER-QUARTER



Quarterly transaction volumes indicate highly concentrated buying sprees during **Q4 (Festive Quarter)**, demonstrating over **32% spike** in average unit value.

PREFERRED PAYMENT BEHAVIOR

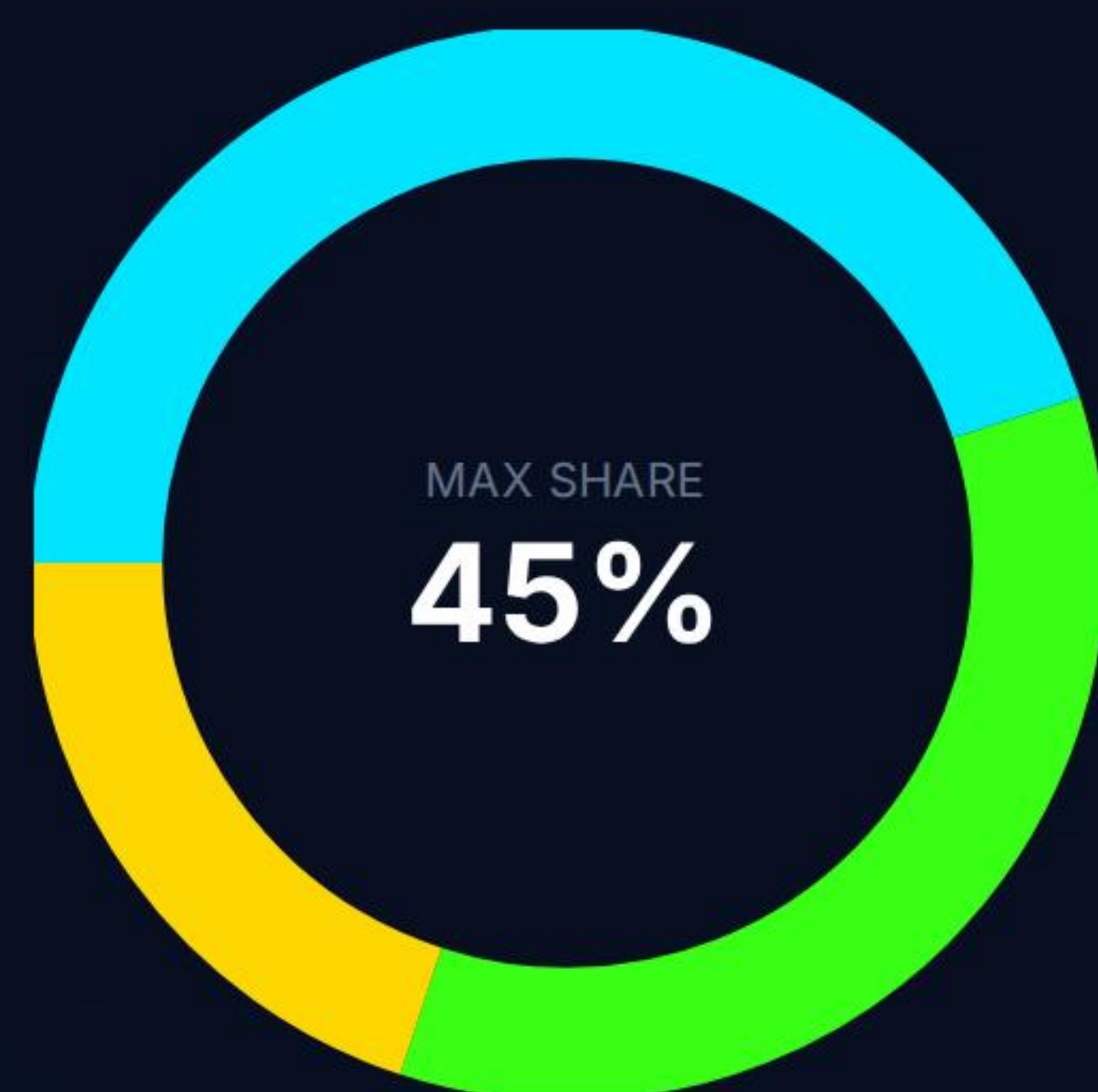
PROPORTIONAL SHARE



Over **85%** of high-value purchases (>\$500) were completed using Digital Payment routes (Credit / UPI).

Dashboard 3: Consumer Demographics

SEGMENTATION



AGE COHORT ANALYSIS

- Gen Z (Ages 18-25): 45% Share of Total Purchases
- Millennials (Ages 26-40): 35% Share of Total Purchases
- Gen X & Baby Boomers (Ages 41+): 20% Share

DAX Engine Calculations

FORMULAS & DAX

DYNAMIC MEASURE FORMULATION

TOTAL REVENUE CALCULATION

$$\text{Total Revenue} = \sum_{i=1}^n (\text{Units Sold}_i \times \text{Price Per Unit}_i)$$

YEAR OVER YEAR GROWTH (YoY)

$$\text{YoY Growth} = \frac{\text{Revenue}_{\text{Current}} - \text{Revenue}_{\text{Prior}}}{\text{Revenue}_{\text{Prior}}}$$

DAX CALCULATIONS APPLIED

- ✓ **Dynamic SumX Iteration:** Explicitly utilizes `SUMX` inside the Semantic Data Model over individual transaction rows to maintain computational speed.
- ✓ **Time-Intelligence Integration:** YoY growth utilizes `SAMEPERIODLASTYEAR` linked with the active `Dim\_Calendar` table for seamless comparisons.
- ✓ **Demographic Buckets:** Age segments generated using `SWITCH` statement dynamically grouping transaction data without duplicating database size.

Strategic Insights & Takeaways

EXECUTIVE SUMMARY



Brand Dominance

Samsung and Apple control over ****68%**** of the total market revenue pool, indicating high brand affinity.



Millennial Velocity

Younger target audiences (Ages 18-30) demonstrate rapid upgrading behavior, average refresh speed of ****14 months****.



Tier-1 Revenue Hubs

Metros like Hyderabad and Mumbai contribute to over ****55%**** of net margins, presenting immediate expansion opportunities.



Payment Corridor

Credit Card usage corresponds directly with a ****15% increase**** in average ticket value compared to offline/COD purchases.

System Roadmap & Cloud Vision

FUTURE SCOPE

- ✓ **AI Forecasting Integration:** Incorporate Machine Learning and Python-based predictive algorithms inside Power BI to forecast demand trends.
- ✓ **Cloud-Based Collaboration:** Host the solution directly on the Power BI Service cloud portal for instant cross-organizational workspace access.
- ✓ **Real-Time Sales Feeds:** Link analytical dashboards directly with ERP databases using Live DirectQuery gateways for minute-by-minute analytics.
- ✓ **Customer Recommendation Engine:** Leverage transactional correlation records to dynamically prompt relevant mobile accessory products.



Cloud Gateway Architecture

Transitioning desktop Semantic Modeling to distributed Enterprise Datasets, ensuring automated schedule-refreshes and mobile dashboard viewing.



Questions & Answers

Interactive platform built using Power BI Desktop, MySQL, and Python. We now open the floor for technical queries and modeling feedback.

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